

Deep Learning for Coronary Artery Disease Detection from SPECT-MPI: A Synthetic Sample Study

A. Sample, B. Example — Synthetic Research Group (fictional)

SYNTHETIC SAMPLE DOCUMENT — created for software demonstration of the Medical RAG Chatbot. The study, data, and results below are entirely fictional and must not be used for any clinical or research purpose.

Abstract. We present a synthetic evaluation of a convolutional neural network for detecting coronary artery disease (CAD) from single-photon emission computed tomography myocardial perfusion imaging (SPECT-MPI). Using a fictional cohort of 1,024 patient studies, a ResNet-50 model trained with transfer learning reached an accuracy of 0.91 and an area under the ROC curve (AUC) of 0.94. Grad-CAM was used to make predictions interpretable for clinicians.

1. Introduction

Coronary artery disease is a leading cause of morbidity worldwide. SPECT-MPI is widely used to assess myocardial perfusion. This synthetic study illustrates how a deep-learning pipeline could support automated CAD screening while remaining explainable.

2. Dataset

The fictional dataset comprises 1,024 SPECT-MPI patient studies, split into 60% training, 20% validation, and 20% test partitions. Each study is labelled as CAD-positive or CAD-negative. Images were resized to 224x224 pixels and intensity-normalised.

3. Methods

We fine-tuned a ResNet-50 backbone pre-trained on ImageNet. The model was trained for 50 epochs with the Adam optimiser at a learning rate of $1e-4$ and a batch size of 32. Data augmentation included rotation, horizontal flipping, and random cropping. Particle Swarm Optimisation was used to tune the classification threshold.

4. Results

On the held-out test set the model achieved an accuracy of 0.91, an AUC of 0.94, a sensitivity of 0.89, and a specificity of 0.92. These fictional numbers are provided only to give the demonstration chatbot concrete, citable facts to retrieve.

5. Explainability

Grad-CAM heatmaps highlighted the myocardial regions that most influenced each prediction, allowing a reviewing clinician to sanity-check the model's focus against known perfusion defects.

6. Conclusion

This synthetic study shows how an explainable ResNet-50 pipeline could assist CAD detection from SPECT-MPI. All figures are fictional and intended solely for software demonstration.